Assignment - 2
(Fourier Series)

1. State Dirichlet's Conditions for expansion of $f(x)$ in fourier Series.

2. Obtain the fourier Series to represent $f(x) = \frac{1}{4}(\pi - x)^2$ in the interval $0 \le x \le 2\pi$

$\boxed{\text{Ans: } f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}}$

Hence obtain the following relations:

(i) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}$

(ii) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}$

3. Expand $f(x) = |x|$ in a fourier series in $(-\pi, \pi)$

$\boxed{\text{Ans} = |x| = \frac{\pi}{2} - \frac{4}{\pi}\left[\cos x + \frac{\cos 3x}{3^2} + \cdots\right]}$

4. Expand $f(x) = x\sin x$, $0 < x < 2\pi$ as a fourier Series.

$\boxed{\text{Ans} = f(x) = -1 - \frac{1}{2}\cos x + \pi\sin x + \sum_{n=2}^{\infty} \frac{2}{n^2-1}\cos nx}$

5. Find fourier series of $f(x) = x^3$ in $(-\pi, \pi)$

$\boxed{\text{Ans} = f(x) = 2\sum_{n=1}^{\infty}\left(\frac{6}{n^3} - \frac{\pi^2}{n}\right)(-1)^n \sin nx}$

6. Find the fourier Series for the function

$$f(x) = \begin{cases} -1 & -\pi < x < -\pi/2 \\ 0 & -\pi/2 < x < \pi/2 \\ 1 & \pi/2 < x < \pi \end{cases}$$

$\boxed{\text{Ans} = f(x) = \frac{2}{\pi}\left[\sin x - \sin 2x + \frac{\sin 3x}{3} \cdots\right]}$

7. Obtain Fourier Series for the function $f(x)$ given by.

$$f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi \le x \le 0 \\ 1 - \frac{2x}{\pi}, & 0 \le x \le \pi \end{cases}$$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$

Change of interval:

8. Find fourier expansion for the function

$f(x) = x - x^2, \quad -1 < x < 1$

$$\boxed{\text{Ans; } f(x) = -\frac{1}{3} + \frac{4}{\pi^2}\left(\frac{\cos \pi x}{1^2} - \frac{\cos 2\pi x}{2^2} + \frac{\cos 3\pi x}{3^2} - \cdots\right) + \frac{2}{\pi}\left(\frac{\sin \pi x}{1} - \frac{\sin 2\pi x}{2} + \frac{\sin 3\pi x}{3} - \cdots\right)}$$

9. Obtain fourier Series for function $f(x) = \begin{cases} \pi x & 0 \le x \le 1 \\ \pi(2-x), & 1 \le x \le 2 \end{cases}$

$$\boxed{\text{Ans} = f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \cdots\right)}$$

10. Develop $f(x)$ in a fourier Series in the interval $(0, 2)$ in $f(x) = \begin{cases} x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$

$$\boxed{\text{Ans; } f(x) = \frac{1}{4} - \frac{2}{\pi^2}\left(\cos \pi x + \frac{\cos 3\pi x}{3^2} + \cdots\right) + \frac{1}{\pi}\left(\sin \pi x - \frac{\sin 2\pi x}{2} + \frac{\sin 3\pi x}{3} + \cdots\right)}$$

Half Range Series

11. Expand $\pi x - x^2$ in a half - range sine series in the interval $(0, \pi)$ upto the first three terms.

$$\boxed{\text{Ans: } \pi x - x^2 = \frac{8}{\pi}\left(\sin x + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \cdots\right)}$$

12. Obtain a half - range cosine series for

$$f(x) = \begin{cases} kx & \text{for } 0 < x < l/2 \\ k(l-x) & \text{for } l/2 < x < l \end{cases}$$

Deduce the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$

13. Obtain the half - range sine series for the function $f(x) = x^2$ in the interval $0 < x < 3$.

Ans: $f(x) = \sum_{n=1}^{\infty}\left(\frac{18}{n\pi}(-1)^{n+1} + \frac{36}{n^3\pi^3}[(-1)^n - 1]\right)\sin\frac{n\pi x}{3}$

Application of Fourier Series

14. A tightly stretched string with fixed end point $x = 0$ and $x = l$ is initially in a position given by $y = y_0 \sin^3(\pi x/l)$. If it is released from rest from this position, find the displacement $y(x,t)$.

$$\boxed{\text{Ans: } y(x,t) = \frac{3y_0}{4}\sin\frac{\pi x}{l}\cos\frac{\pi c t}{l} - \frac{y_0}{4}\sin\frac{3\pi x}{l}\cos\frac{3\pi c t}{l}}$$

15. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is vibrating by giving to each of its points a velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from one end at any time t.

$$\boxed{\text{Ans}; \quad y = \frac{8\lambda l^3}{c\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2m-1)^4} \sin\frac{(2m-1)\pi x}{l} \sin\frac{(2m-1)\pi ct}{l}}$$